Web Scraping Exercise: Reading the Data

Explore web scraping techniques by extracting tables from a Wikipedia page about the world's largest banks. Learn to send HTTP requests, parse HTML responses with BeautifulSoup, and store the extracted data in pandas DataFrames, preparing it for further analysis.

Time to practice and experiment with web scraping. In this exercise, you'll extract data from a Wikipedia page about the world's largest banks.

The web page contains two tables ranking the largest banks in the world. The first table ranks banks by total assets.

List of largest banks

16 languages

Article Talk Read Edit View history Tools

From Wikipedia, the free encyclopedia

The following are lists of the largest banks in the world, as measured by market capitalization and total assets.

By total assets [edit]

This list is based on the April 2023 S&P Global Market Intelligence report of the 100 largest banks in the world. The ranking was based upon assets as reported and was not adjusted for different accounting treatments.[2] Another publication which compiles an annual list of the world's largest banks is *The Banker* magazine. It publishes a list of the World 1000 Largest Banks every July.[3] The financial data published by the July yearly issue of *The Banker* are much more extensive compared to the S&P Top 100 banks, but it is not a publication intended for the general public. Some banks were added due to their annual reports.

Industrial and Commercial Bank of China, the largest bank in the world by total assets

Accounting treatment affects the assets reported: for example, the United States uses US GAAP (as opposed to IFRS), which only reports the net derivative position in most cases, leading to US banks having fewer derivative assets than comparable non-US banks.[4] If JPMorgan Chase reported under the IFRS, it would be ranked 2nd on the list as of 2023, rather than 5th.[2]

Rank	Bank name	Total assets [hide] (2022) (US$ billion)
1	Industrial and Commercial Bank of China Limited	5,742.86
2	China Construction Bank	5,016.81
3	Agricultural Bank of China	4,919.03
4	Bank of China	4,192.12
5	JPMorgan Chase	3,736.62

Wikipedia page: List of largest banks—By total assets

Ask

The second table ranks countries by the number of banks per country.

Banks by country or territory [edit]

Number of banks in the top 100 by total assets [2] [hide]

Rank	Country	Number
1	China	20
2	United States	11
3	Japan	8
4	Germany	7
5	France	6
	Canada	6
	South Korea	6
8	Australia	5
	United Kingdom	5
10	Spain	3
	Netherlands	3
	Singapore	3
	Brazil	3

Wikipedia page: List of largest banks—Banks by country or territory

Your task is to scrape the two tables, store them in a pandas DataFrame, and export them as a CSV file for later analysis.

> **Note:** Fill in the missing code in the following code widgets to complete the exercise.

Send a `get` request

Ask

First, use `requests` to send a `get` request to Wikipedia. After grabbing the response, fetch the response code and verify that it is equal to `200`.

main.py Python 3.10.4

```
# Import
import requests

URL = "https://en.wikipedia.org/wiki/List_of_largest_banks"

# Send a GET requst to "https://www.example.com"
response = requests.get(URL)

# Grab the response status code
<<InsertHere>>

# Check status code and print a message accordingly
<<InsertHere>>
```

Run

Send a get request

Create a Beautifulsoup object

After verifying that the response is successful, import `BeatifulSoup` and parse the response using the `html.parser`. Name the object `soup` in the code.

main.py Python 3.10.4

```
# Parse the response using BeautifulSoup and html.parser
print('Parsing the response.')
soup = <<InsertHere>>

# Print the HTML response
<<InsertHere>>
```

Ask

Run

Create a BeautifulSoup object

Find all wiki tables on the web page

Next, find all the `table` elements on the web page of the class `wikitable`.

main.py Python 3.10.4

```
# Find all objects of class: wikitable
print('\nSearching for HTML objects of type: wikitable')
wiki_tables = soup.<<InsertHere>>(name='table', attrs={'class':'wikitable

# Verify that there are two tables
number_of_tables = <<InsertHere>>
print(f'Found {number_of_tables} tables')
```

Run

Extract all wiki tables

Complete solution code

Here's the complete solution for the above tasks:

main.py Python 3.10.4

```
# Import
import requests
from bs4 import BeautifulSoup

URL = "https://en.wikipedia.org/wiki/List_of_largest_banks"

# Send a GET requst to "https://www.example.com"
```

Ask

```
response = requests.get(URL)

# Grab the response status code
response_code = response.status_code
```

Run

Solution code

- **Lines 2–3:** We import the libraries `requests` and `BeautifulSoup`.
- **Line 8:** We send a `get` request to "https://en.wikipedia.org/wiki/List_of_largest_banks".
- **Line 11:** We fetch the response code.
- **Line 14:** We print a message accordingly if the response code indicates a successful request.
- **Line 21:** We create a `Beautifulsoup` object named `soup`. We use `html.parser` to parse the response.
- **Line 25**: We find all tables on the webpage of the class `wikitable`.
- **Line 28:** We check the length of the `BeautifulSoup` object `wiki_tables`.

Mark As Completed

Ask